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Experiment No 2

Aim: Map ER / EER to Relational schema model

Theory:

conversion from ER diagrams to tables

step1 :- Mapping of Regular Entity types:

① For each regular (strong) entity type E in the ER schema create a Relation R that includes all the simple attributes of E. Include only the simple component attributes of a composite, the set of simple attributes that form it will together form the primary key of R

step2 :- Mapping of weak Entity Types

① for each weak entity type W in the ER schema with owner entity type E, create a relation R and include all the attributes of W and the specific attributes of R. In addition include the primary key attributes of the owner entity type. The primary key of R is the combination of the primary key(s) of the owner(s) and the partial key of the weak entity type W.

step 3 :- Mapping of Binary 1 : 1 Relationship types.

for each regular binary 1:1 relationship type R in the

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ER schema, identify the relations S and T that correspond to the Entity Types participating in R. Choose one of the relations - S, say - and include as a foreign key in S the primary key of T. It is better to choose an entity type with total participation in R in the role of S. Include all the simple attributes (or simple components of a composite attributes) of the 1:1 relationship type R as attributes of S.

Step 4: Mapping of Binary 1:N relationship types.

For each regular binary 1:N relationship type R, identifying the relation S that represent the participating Entity type at the N side of the relationship. Include as foreign key in S the primary key of T that represents the other entity type participating in R; this is done because each entity instance of the N side is related to at most one entity instance of the 1-side of the relationship type. Include any simple attributes (or simple components of composite composite attributes) of the 1:N relationship type as attributes of S

Step 5: Mapping of Binary M:N relationship Types

For each binary M:N relationship Type R, create a new relation S to represent R, include the primary Key

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of the participating entity types; their combination will form the primary key of S. Also include any simple attributes of the M:N relationship types (or simple components of a composite attributes) as attributes of S.

step 6: Mapping of Multivalued attributes:

For each Multivalued attribute A, create a new relation R. This relation R will include an attribute corresponding to A, plus the primary key attribute K - of the relation that represent the entity type or relationship type that has A as an attribute. The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.

step 7: Mapping of N-ary Relationship types:

For each N-ary relationship type R, where N > 2, create a new relation S to represent R. Include the primary keys of the relationship that represent the participating entity types. Also include any simple attributes of the N-ary relationship types (or simple components of a composite attributes) as attributes of S.

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Mapping EER model constructs to Relations

Step 8: options for Mapping Specialization or Generalization

Convert each specialization with m subclasses {S1, S2, ..., Sm} and generalized superclass C where the attributes of C are {k, a1, ... an} and k is the (primary) key, into relational schemas using one of the four options:

option 8A :- Multiple relations - superclass and subclasses

Create a relation L for C with attributes Attrs(L) = {k, a1, ... an} and PK(L) = k. Create a relation Li for each subclass Si, 1<i<m with the attributes Attrs(Li) = {k} ∪ {attributes of Si} and PK(Li) = k. This option works for any specialization (total or partial, disjoint or overlapping).

option 8B :- Multiple relations - Subclass relations only

Create a relation Li for each subclass Si, 1<i<m, with the attributes Attr(Li) = {attributes of Si} ∪ {k, a1, ..., an} and PK(Li) = k. This option only works for a specialization whose subclasses are total (every entity in the superclass must belong to (at least) one of the subclasses

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If the specialization is overlapping an entity may be duplicate in several relations.

Option 8C: single relation with one type attribute

create a single relation L with attributes Attrs(L) = {K, a1, ..., an} U of attributes of S1} U .... U {attributes of Sm} U {t} and PK(L) = k. the attribute t is called a type (or discriminating) attribute that indicates the subclass to which each tuple belongs (Disjoint)

Option 8D: single relation with multiple type attributes

create a single relation schema L with attributes Attrs(L) = {K, a1, ... an} U {attributes of S1} U .... U {attributes of Sm} U {t1, t2, ... tm} and PK(L) = k. each ti, 1 < i < m, is a Boolean type attribute indicating whether a tuple belongs to the subclass Si (overlap)

Step 9: Mapping of Union Types (categories)

- For mapping a category whose defining superclasses have different keys, it is customary to specify a new key attribute, called a surrogate key, when creating a relation to correspond to the category.

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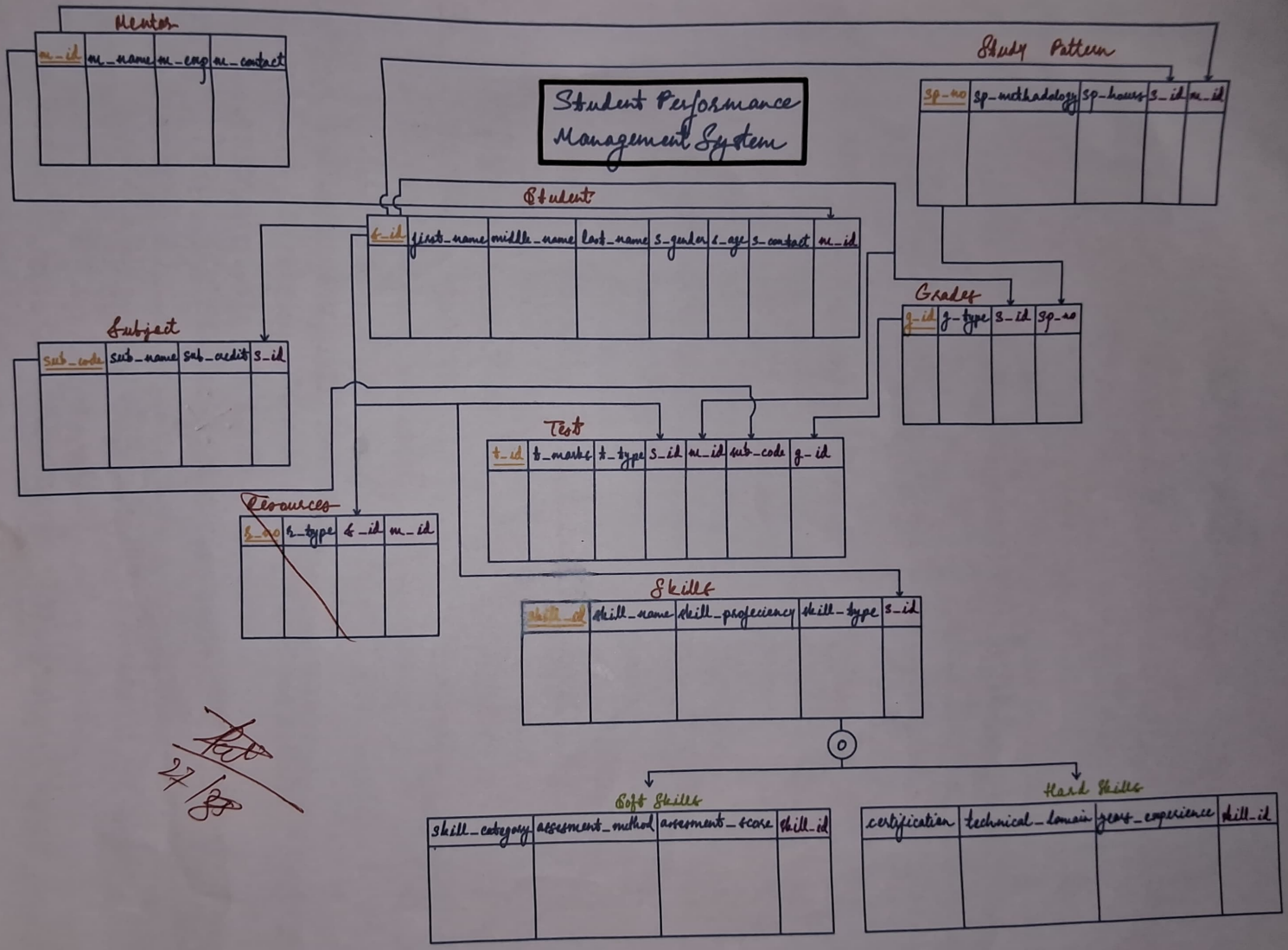
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- For Example we can create a relation owner to correspond to the owner category and include any attributes of the category in this relation

Conclusion:

Thus we have studied the mapping of ER to relational table.

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Student Performance Management System
Mentor
m_id
m_name
m_exp
m_contact
Study Pattern
sp_no
sp_methodology
sp_hours
s_id
m_id
Student
s_id
first_name
middle_name
last_name
s_gender
s_age
s_contact
m_id
Subject
sub_code
sub_name
sub_credit
s_id
Grades
g_id
g_type
s_id
sp_no
Test
t_id
t_marks
t_type
s_id
m_id
sub_code
g_id
Resources
r_no
r_type
t_id
m_id
Skills
skill_id
skill_name
skill_proficiency
skill_type
s_id
Soft Skills
skill_category
assessment_method
assessment_score
skill_id
Hard Skills
certification
technical_domain
years_experience
skill_id